

Homework 6 File Submission

- A. Find a formula for $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{2^n}$ by examining the values of this expression for small values of n . Then use mathematical induction to prove the formula.
- B. Prove that $3^n < n!$ if n is an integer greater than 6.